USER GUIDE

Midiplus UP
Bitwig Controller Script

Modes, button map, jog-wheel combos, and setup

Script file	`MidiplusUP-MCU.control.js`
Hardware mode	Standard MCU mode (manual sections 3.3 / 8)
Bitwig API	Version 25+ - Bitwig Studio 6.x (confirmed 6.0 / 6.1)
Version	Starcycle + Claude + 3.0.0-native-faders
Repository	starcycle-controller/bitwig-midiplus-up

Contents

1. Intended Use Case

This script turns a single Midiplus UP unit (in standard MCU mode) into a dedicated Bitwig mixing and gain-staging control surface: 8 motorized channel strips plus a master strip, not a multi-unit MCU array.

It is built around three everyday jobs:

- **Mixing** - motorized-fader control of track Volume and Pan, with fader snap points (to true zero, and to the round dB marks printed on the hardware) so you can land exactly where you mean to.
- **Gain-staging** - a dedicated mode that points the same faders/encoders at a per-track utility device instead of the track's own Volume/Pan, for a separate, non-destructive gain-staging pass ahead of the main mix (see Setup below).
- **Plugin control and navigation** - encoder control of a selected device's 8 macros, jog-wheel scrubbing/zooming/clip and track navigation, scene launching, and automation-mode cycling - all without reaching for the mouse.

On top of that, the script adds a set of quality-of-life extras: whole-project mixer snapshots (store/recall a full Volume+Pan balance across every track, not just the visible 8), auto-banking (the bank follows track selection made elsewhere in Bitwig), and a fader position self-test.

2. One-Time Setup

> Takeover Mode doesn't matter for this controller
> Bitwig's **Takeover Mode** (Settings -> Controllers - Pick Up / Jump / Value Scaling) is Studio-wide, applying to every connected controller - there's no per-controller override in Bitwig 6.1. This script only uses Bitwig's native binding system for the 8 motorized faders, and each one opts out of Takeover Mode already (the motor keeps the fader in sync directly). Everything else is raw MIDI handled by the script's own code, bypassing Bitwig's Takeover pipeline entirely. **So this setting changes nothing about how the Midiplus UP behaves** - pick it based on your other connected gear instead. As a rule of thumb: **Pick Up (Catch)** is safer for non-motorized controllers; **Jump (Immediate)** suits motorized ones that already track their own position, like this one.

Hardware mode

In Bitwig, go to **Settings -> Controllers**, add this script, and confirm the Midiplus UP is switched to standard **MCU mode** (hardware manual sections 3.3 / 8) - not one of the Logic/Cubase/Live "customized" modes.

> After loading, check that Bitwig's Controllers panel (or Controller Preferences -> About -> Version) shows the version string on the cover page. An older string means Bitwig is still running a cached copy of a previous version of the script.

The Ableton Live overlay

This setup deliberately leaves the plastic Ableton Live overlay's **left and right** pieces attached, and removes only its **top** piece. That matters for reading the button map in this guide without confusion:

- Buttons under the **top** piece show their real, printed MCU labels - TRACK/IO, SEND, PAN, PLUG-INS, RETURNS, BANK/CHANNEL, F1-F8, the four modifier buttons, the jog-wheel section, and transport. These are the labels used throughout this guide.
- Buttons under the **left and right** pieces still show old Live-mode stickers (SESS/ARR, CLIP/FX, BROWSER, DETAIL, LOOP, PUNCH OUT, HOME, END, and others) that have nothing to do with what the button actually does in this script. Wherever the button map below shows *"(Live label: X)"*, that is this case - ignore the sticker, the Bitwig behavior listed in that row is what actually happens.

Confirmed on hardware: switching the unit between Live mode and standard MCU mode does not change any button's note number - only local LED/display behavior differs - so this mapping holds regardless of which overlay pieces happen to be attached.

The TRLVL device (required for Gain-Staging mode)

> The **PAN** button's Gain-Staging mode (see Modes below) does not control the track's own Volume/Pan - it controls a separate utility device on that track's chain. For this to work, that device **must exist and must be named exactly `TRLVL`** (case-sensitive, no extra text).
>
> Bitwig's built-in **Tool** device (browser category **Audio FX**) is the natural fit: its first two native parameters are already Gain and Pan, exactly what this script expects to find on that device's first Remote Controls page.
>
> **To set it up on a track:**
>
> 1. Add a **Tool** device anywhere in that track's device chain (a common choice is right before the track's own output, after EQ and other corrective processing).
> 2. Rename it to exactly `TRLVL`.
> 3. Repeat per track you want to gain-stage this way.
>
> If you press PAN on a track with no `TRLVL` device yet, the script opens the browser at the end of that track's chain automatically, so you can add one in a couple of clicks.

3. Modes at a Glance

Mode	How to enter	What the faders / encoders control
Mixer (default)	TRACK/IO button, or leaving any other mode	Faders: track Volume. Encoders: track Pan. FLIP swaps faders/encoders between the two.
Sends	SEND button (cycles Sends 1-8 → 9-16 → back to Mixer, or 1-8 ↔ Mixer if Send/Return Bank Size = 8); SHIFT+SEND jumps straight to Sends 9-16	Faders/encoders: the selected track's send levels.
Device / Plugin	PLUG-INS button, or F1-F8 (default orange state, selects device 1-8 directly and opens its window)	Encoders: always the 8 device Remote Control macros. Faders: Volume by default, the same macros once FLIP is on.
Gain-Staging (Tool Volume)	PAN button, toggle - a sub-mode of Mixer	Faders/encoders: the Gain/Pan of that track's `TRLVL` device instead of its own Volume/Pan - requires the Setup step above.
Returns	RETURNS button	Same as Mixer/Sends, but the 8 channel strips are the Return tracks instead of Main tracks.
Scene	B.T.A. button	Shows the clip launcher and switches Bitwig to the Mix panel layout; the jog wheel selects/launches scenes instead of scrubbing the timeline.

Every mode-changing button fully resyncs the mode LEDs, LCD text, channel-strip LEDs, and fader/encoder bindings together in one step, and automatically closes a Device-mode plugin window when leaving that mode - jumping directly between any two modes always lands in one consistent state.

4. Modifier Buttons at a Glance

`SHIFT` `OPTION` `CTRL` `ALT` - the four buttons directly under the F-keys - are held modifiers for most of what this controller does. Encoder turns, jog-wheel turns, and some button taps all change behavior depending on which is held. A quick tap-and-release of any of them (without using it to modify anything else) also has its own configurable standalone action, described in the full settings reference (see Section 9).

Modifier	Broad role
`SHIFT`	Stepped/coarser adjustment on encoders; jump-to-start/end on bank scrolling; "store" actions (Mixer Snapshots, F-key green state) and the automation-write-enable toggle on DRAW.
`OPTION`	Loop-length halve/double on the wheel; "recall" actions (Mixer Snapshots) and hover-lock for last-clicked-parameter adjustment.

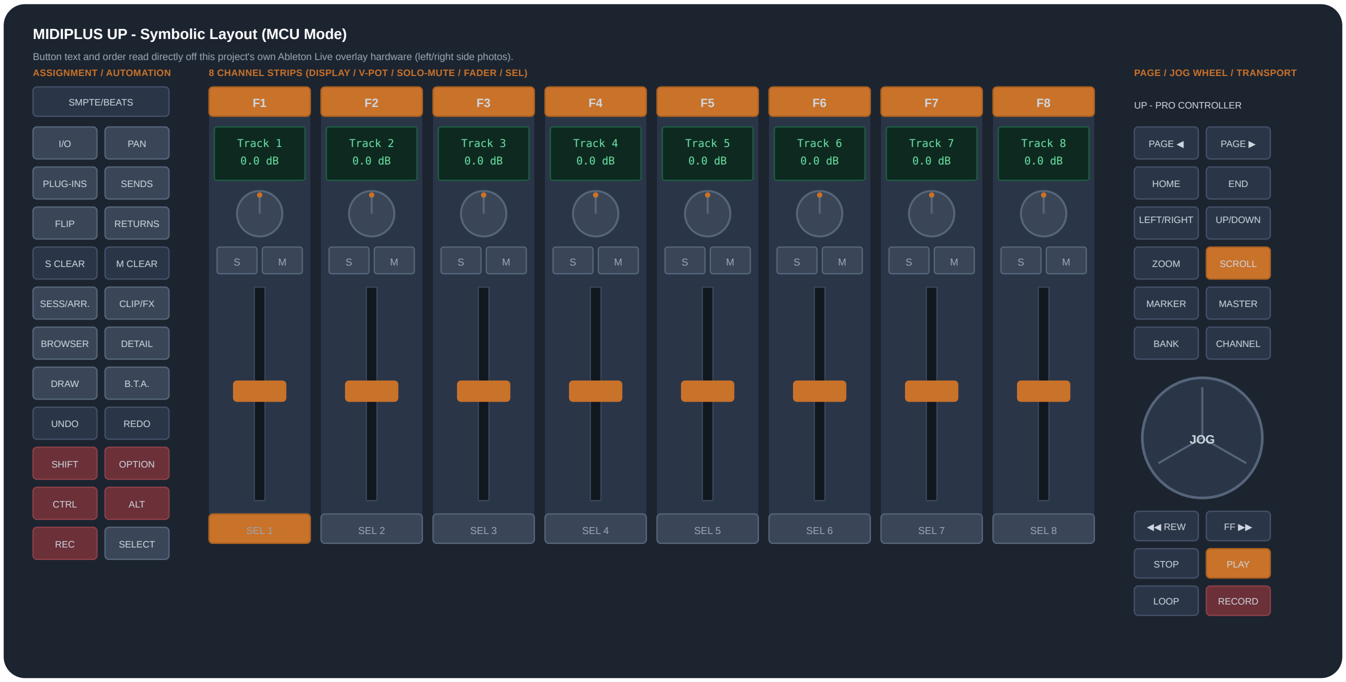
Modifier	Broad role
CTRL	The most-used combo modifier: steps devices, steps clip/track selection, and combines with SHIFT/ALT for clip/track duplicate-or-delete gestures.
ALT	Fine-resolution adjustment on encoders/faders; adjusts the last-clicked Bitwig GUI parameter via the wheel; combines with F8 for the Fader Position Test.

5. Modes in Detail

- **Mixer** (default) - faders/encoders control track volume/pan, or a `TRLVL` device's Gain/Pan if Gain-Staging mode is active (PAN button).
- **Sends** - faders/encoders control the focused track's sends. SEND cycles 1-8 → 9-16 → back to Mixer by default, or 1-8 → Mixer directly if **Send/Return Bank Size** is set to 8. SHIFT+SEND always jumps straight to sends 9-16.
- **Device** - encoders *always* control the selected device's 8 Remote Control macros, regardless of FLIP. Faders control track volume by default and swap to the macros when FLIP is on (press again to revert) - FLIP only affects the faders here, not the encoders. Entered via PLUG-INS, or via F1-F8 in their default/orange state, which also jump directly to device 1-8 on the chain and open its plugin window. Pressing the same F-key again for the already-selected device toggles its window closed/open instead of reselecting it.
- **Scene** - entered via B.T.A.: shows the clip launcher, switches Bitwig to the Mix panel layout, and the jog wheel selects/launches scenes instead of scrubbing.

FLIP swaps faders and encoders between volume and pan in Mixer mode, and faders only between volume and macros in Device mode.

6. Confirmed Button Map



Original schematic diagram - button text and top-to-bottom order read directly off this project's own Ableton Live overlay hardware (photos of the left and right control columns), not the generic manual.
Exact note numbers and Bitwig behavior are in the Confirmed Button Map (README.md / User Guide Section 6).

Original schematic diagram for this project - button text and order read directly off this project's own Ableton Live overlay hardware. Not a photo; exact note numbers are in the table below.

Live-tested against real hardware in MCU mode. Switching the hardware between Live mode (with the overlay) and standard MCU mode does not change any button's note number - only local-firmware LED behavior differs between modes, so this map holds in either.

Notes	Function	Bitwig behavior
0-7	Rec Arm 1-8 (SEL row while the standalone REC button is toggled on)	Toggle record-arm
8-15	Solo 1-8	Toggle solo
16-23	Mute 1-8	Toggle mute
24-31	Select 1-8 (double-press folds/unfolds a group track)	Select in mixer / toggle group expand
32-39	Encoder push-click	Center pan (Mixer) / reset send (Sends) / Device: hold+turn = fine resolution, or reset macro, or open/close plugin window

Notes	Function	Bitwig behavior
40	TRACK/IO	Toggle Track Inspector, or switch to Mixer mode
41	SEND	Sends mode toggle (SHIFT = jump to Sends 9-16)
42	PAN	Toggle TRLVL tool-device Gain/Pan control (Gain-Staging mode)
43	FLIP	Swap faders/encoders between volume and pan (Mixer) or volume and macros (Device)
44	PLUG-INS (SHIFT = EQ Mode)	Toggle Device mode. SHIFT+PLUG-INS jumps to the last EQ in the chain; pressed again, exits to Mixer
45	RETURNS	Swap channel strips to/from the Return Tracks bank
46	BANK PREV	Page track bank back (SHIFT = jump to first)
47	BANK NEXT	Page track bank forward (SHIFT = jump to last)
48	CHANNEL PREV	Nudge one channel back (CTRL = prev device / tempo down)
49	CHANNEL NEXT	Nudge one channel forward (CTRL = next device / tempo up)
50	UNDO	Undo
51	REDO	Redo
52	NAME/VALUE	Unbound (no Bitwig equivalent)
53	SMPTE/BEATS	Hardware-only: toggles F1-F8 backlight/range; not bound in Bitwig
54-61	F1-F8 (default/orange state)	Select device 1-8 directly and open its window; pressing the already-selected key again toggles the window
62-69	F1-F8 (green state, via SMPTE/BEATS)	Configurable editing function per key (defaults: F1=Duplicate, F2=Consolidate, F3-F8=None)
70-73	SHIFT / OPTION / CTRL / ALT	Modifier hold state; standalone tap is configurable
74	(Live label: SESS/ARR)	Toggle clip launcher / arranger view
75	(Live label: CLIP/FX)	Toggle device / clip view
76	DRAW	Plain: cycle automation write mode (Latch/Touch/Write). SHIFT: toggle Automation Write on/off. OPTION: show/hide automation lanes
77	(Live label: BROWSER)	Toggle browser panel
78	(Live label: DETAIL)	Toggle note/automation editor panel
79	B.T.A. (ALT+press = toggle metering plugin window)	Toggle Scene mode; ALT+press toggles the metering plugin's window, independent of mode - see README's Master Wheel section

Notes	Function	Bitwig behavior
80, 81, 87	Unconfirmed	Unbound - needs testing
82	PAGE (left arrow)	Device mode: page macro bank back. Mixer mode: jump to previous cue marker and follow it with the loop
83	PAGE (right arrow)	Device mode: page macro bank forward. Mixer mode: jump to next cue marker and follow it with the loop
84 / 85	-	Jump to previous / next cue marker
86	(Live label: LOOP)	Toggle arranger loop
88	(Live label: PUNCH OUT)	Toggle punch-out (CTRL = set loop end from playhead)
89	(Live label: HOME); SHIFT = add "Bar N" cue marker	Jump playhead to project start; SHIFT+HOME adds an auto-named cue marker
90	(Live label: END)	Jump playhead to loop start
91-95	Transport: REWIND / FF / STOP / PLAY / RECORD	Standard transport
96 / 97	Cursor UP / DOWN	Arrow key, or zoom track height (while ZOOM toggled)
98 / 99	Cursor LEFT / RIGHT	Arrow key, device select prev/next in Device mode, or zoom timeline (while ZOOM toggled)
100	ZOOM	Toggle zoom mode for cursor arrows
101	Jog wheel push	Momentary "Pan Mode" hold; ALT+press = select item at playhead; SHIFT+CTRL+press = same, one-shot; launches selected scene in Scene mode
104-111	Fader touch 1-8	Optionally selects that channel's track
112	Fader touch (Master)	Optionally selects the master track - not confirmed on this 8-fader unit
CC 16-23	Rotary encoders 1-8	Mode-dependent (pan/send/macro); SHIFT = stepped or fine adjust
CC 60	Jog wheel	Arranger scrub, or bar/loop/tempo nudge with modifiers, or scene navigation in Scene mode
Pitch bend ch 0-7	Faders 1-8	Motorized fader input/output, native hardware binding
Pitch bend ch 8	Master (jog wheel, MASTER mode)	No separate master fader on this unit - the jog wheel substitutes pitch-bend on this channel instead; parsed manually by the script rather than a native binding, see note below

7. Jog-Wheel "Mode" Buttons

This hardware has its own local mode buttons for the jog wheel (CURSOR / SCROLL / ZOOM / MASTER / MARKER / NUDGE / BANK / CHANNEL, per the manual's "Multi-Purpose Jog Wheel Section"). Most send no MIDI at all - they are purely local firmware state that changes what the wheel itself sends when subsequently turned.

Mode	Mode button itself	Wheel click	Wheel turn
SCROLL (base/default)	No MIDI at all	Note 101	CC 60 (normal scrub)
ZOOM	Note 100 (toggle)	Nothing	Note-On 96/97 (up/down) or 98/99 (left/right)
MARKER	No MIDI at all	Nothing	Note-On 84/85 - previous/next cue marker
BANK	Note 46/47	Nothing	Note-On 46/47
CHANNEL	Note 48/49	Nothing	Note-On 48/49
MASTER	No MIDI at all	Not yet tested	Pitch-bend, channel 9 - drives master volume, see note below

(CURSOR and NUDGE modes not yet tested.)

> **This unit has no separate physical master fader** - MASTER mode substitutes the jog wheel for one. Confirmed with an independent OS-level MIDI monitor (`receivemidi` , reading the hardware port directly, bypassing this script entirely): turning the wheel with MASTER engaged sends absolute pitch-bend on MIDI channel 9. Unlike the 8 track faders (which use a native Bitwig hardware binding, invisible to the script's own code), this channel is read and interpreted manually by the script itself, so it drives master volume by default but can also be repurposed to open/close a metering plugin's window instead, without ever touching master volume while that's active - see the README's **Master Wheel** Controller Preferences section.

8. Jog Wheel Modifier Combos

CRITICAL - THE WHEEL-MODE SELECTOR MUST BE ON SCROLL

Every combo in this section requires the hardware's own wheel-mode selector (see Section 7) to be set to **SCROLL**, the base/default mode. The physical jog wheel only sends CC 60 - the message every combo below is built on - while that local firmware mode is active. Switch it to ZOOM, MARKER, BANK, or CHANNEL and the wheel repurposes its MIDI output entirely (Note-On messages instead of CC 60), so none of the combos below will fire, with no error or feedback to explain why. If a wheel combo suddenly "stops working," check the wheel-mode selector first.

In priority order (each returns before the next is checked, so only one applies per turn): Scene mode active (move the scene cursor) > **SHIFT** + **CTRL** (configurable) > **ALT** + **CTRL** (configurable) > **CTRL** alone (Device mode: step devices; otherwise: select next/previous arranger clip/item) > **SHIFT** + **ALT** (nudge the selected arranger item) > **ALT** alone (adjust the last-clicked GUI parameter) > PLUG-INS held (step devices) > BANK held (page remote-control pages) > **OPTION** alone (halve/double loop length) > **SHIFT** alone (shift loop by a bar) > default (scrub by whole bars).

SHIFT+CTRL and ALT+CTRL + Jog Wheel (turn)

Each independently runs whichever action its own Controller Preferences dropdown is set to - freely invertible, both offer the same 5 options:

- **Scale Clip Size** - right doubles the selected clip's content, left halves it.
- **Duplicate/Delete Clip** - right duplicates the selection, left deletes it.
- **Duplicate Clip** (SHIFT+CTRL's default) - right duplicates; left is always a no-op.
- **Duplicate/Delete Track** (ALT+CTRL's default) - right duplicates the current track, left deletes it.
- **Duplicate Track** - right duplicates the current track; left is always a no-op.

Turning left in either "Duplicate/Delete" option is gated by a separate "Allow Delete (Turn Left)" setting (default on) - off, only duplicate (right) is live. The plain Duplicate options never need that toggle, since turning left there is always a no-op by design.

CTRL + Jog Wheel (turn, outside Device mode)

Selects the next/previous arranger clip/item. With a clip selected, steps between clips on that track; with nothing selected, steps track-to-track instead. In Device mode, CTRL + Jog Wheel instead steps through devices on the current chain.

> **Click a clip with the mouse first to step between clips with this combo.** There is currently no button/wheel gesture on this hardware that can select ("anchor") a clip from nothing - every generic Bitwig action tried for that was confirmed on hardware to do nothing at all (see Section 9). Once a clip is selected by clicking it, CTRL+Wheel reliably steps to the next/previous one from there - it just cannot establish that starting point on its own.

OPTION + Jog Wheel

Halves (turn left) or doubles (turn right) the arranger loop length, capped at 256 bars doubling, floored at 1 whole bar halving.

ALT + Jog Wheel

Adjusts whatever parameter was last clicked in Bitwig's own GUI - click any knob/slider/fader once in Bitwig, then hold ALT and turn the wheel to dial it in without touching the mouse again.

ALT + Jog Wheel Press attempts to select whatever's at the current GUI focus - the same underlying action confirmed non-functional for clip selection elsewhere in this guide, so treat this as likely not actually selecting anything rather than a reliable click substitute.

SHIFT+ALT + Jog Wheel nudges whatever's currently selected in the arranger left/right by one grid step per message - this part does work once something is already selected. Since the wheel-press "select" step above is unreliable, **click the item with the mouse first** (same as CTRL+Wheel above), then hold SHIFT+ALT and turn the wheel to "drag" it.

OPTION + Jog Wheel Press toggles a hover-lock: locks the ALT+wheel parameter-adjust combo onto whatever parameter the mouse is currently hovering over, without needing an exact click first.

Default (no modifier) Jog Wheel

Always moves in whole bars, anchored on a bar start, at a configurable number of bars per accumulated wheel-tick threshold. An optional adaptive mode can instead scale the jump with the current Arranger zoom level, so a tick always covers roughly the same visual distance regardless of zoom.

9. Where to Find Everything Else

This guide covers usage: modes, the button map, and the jog-wheel combos. The project's `README.md` (in the repository root) covers the rest, including:

- Every configurable Controller Preferences setting, grouped by category (Plugin Mode, Function Keys, Zoom, Encoders, Mixer, Debug), with every default value and range.
- Mixer Snapshots (whole-project store/recall) in full detail.
- Development notes and hardware-debugging findings, for anyone extending this script.

`feature-requests/BITWIG-API-FEATURE-REQUESTS.md` (in the repository's `feature-requests/` folder, alongside the Midiplus support email draft) lists the real Bitwig Controller API gaps this project had to work around, for anyone interested in the engineering background.